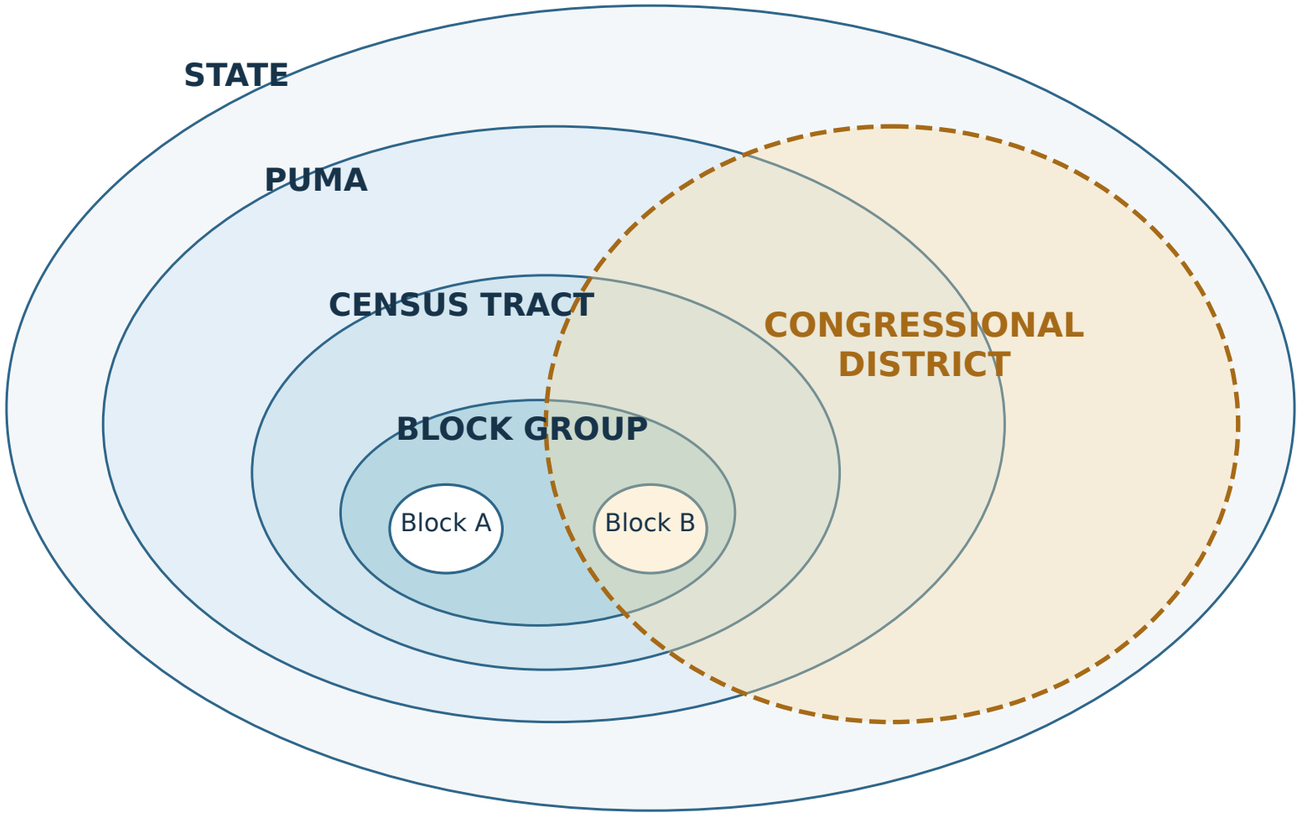
First, how the geographic areas fit together

How 2020 Census statistical areas fit within a state and overlap a 120th Congress district. Shapes are schematic.



Block A is outside the illustrated district. Block B is inside it.

A district can cut across a block group, tract and PUMA.

The file assigns one district code per block. Census tabulates Colorado’s split block in CO-08.

Block

Smallest geographic unit used here; supplies 2020 adult counts.

Block group (BG)

Contains blocks. This is the smallest level with our published CVAP estimates.

Census tract

Contains block groups. The tract lookup locates blocks in 2020 PUMAs.

PUMA

Public Use Microdata Area: a group of whole tracts. Our individual microdata identify this area, not a block group.

Congressional district (CD)

The destination whose probability we calculate within each PUMA.

State

Contains the PUMAs and districts. State codes are part of their unique identifiers.

County, county equivalent and town: a second set of boundaries

Tracts also belong to counties or county equivalents. A PUMA can cross county boundaries, so county does not sit between PUMA and state in the nesting above. In Connecticut, newer ACS identifiers use nine planning regions as county equivalents instead of the eight old counties.

Connecticut towns (county subdivisions) appear only in the relationship file that helps connect old block groups to new planning-region identifiers. They are not an additional level at which we allocate people. The Connecticut conversion is explained on panel 3.

CVAP = citizen voting-age population: citizens aged 18+. ACS = American Community Survey.

Four inputs answer four different questions

Read leading zeros as part of an identifier. ACTUAL rows are verified examples; ILLUSTRATIVE rows explain the calculation.

A Which district contains each block?

Census BEF means Block Equivalency File. It assigns original 2020 blocks to the pinned 120th Congress district plan. CDFP is the district code. ZZ means no district is assigned; it does not itself mean zero population.

ACTUAL: first three rows of NationalCD120.txt

GEOID	STATEFP	CDFP
010010201001000	01	06
010010201001001	01	06
010010201001002	01	06

B Which PUMA contains each tract?

The Census tract-to-PUMA table supplies PUMA5CE. A block GEOID begins with state + county + tract, so its first 11 digits identify the tract. Combine STATEFP with PUMA5CE to identify a PUMA uniquely.

ACTUAL: 2020 Census tract-to-PUMA lookup

STATEFP	COUNTYFP	TRACTCE	PUMA5CE
01	001	020100	01700
01	001	020200	01700
01	001	020300	01700

C How many adults lived in each block?

PL stands for Public Law: P.L. 94-171 is the law behind this Census redistricting data release. P0030001 counts adults aged 18+, including noncitizens. POP100 counts residents of all ages. The code calls these vap20 and pop20.

ILLUSTRATIVE: invented blocks b1–b3, after population matching

GEOID	POP100 → pop20	P0030001 → vap20
b1	90	60
b2	210	140
b3	150	100

D How many citizen adults are estimated?

The 2020–2024 ACS CVAP table supplies one Total estimate per block group. We use lnnumber = 1. This is a survey estimate for the block group; citizen-adult counts are not observed for individual blocks.

ILLUSTRATIVE: three block groups used in panels 3, 5 and 7

geoid / bg_cvap	lnnumber	lntitle	cvap_est
G1	1	Total	120
G2	1	Total	80
G3	1	Total	100

The all-age counts check the joins: block sums match state totals; no population is lost or duplicated;
adult counts lie between zero and all-age counts. The allocation denominator uses adults, not all ages.

Join the geography, then attach one CVAP estimate

A + B + C → block-group/PUMA/district portions → reconcile Connecticut IDs → attach CVAP from D.

Combine A, B and C into portions

Join blocks to PUMAs and district codes, then add adult counts within each block-group / PUMA / district combination. “Coverage” means which places the records map to; it does not mean a survey response rate. Matching checks ask whether every identifier finds the required row.

Connecticut: translate identifiers carefully

Old block group 090093431011 overlaps two newer planning regions. One overlap covers 3,606,413 m²; the other covers 160 m². Replacing the county part of its identifier produces the two candidates shown.

The Census 2024 block-group geographic data contain the first ID, but not the second. We map the whole old group to the first ID and keep every original block. This is a specific boundary approximation, not a rule that always chooses the largest area.

One group estimate must be counted once

After Connecticut reconciliation, attach the published estimate to each group. G1 appears in two district rows, so its source value 120 repeats after the join. Those repeated values are the same estimate, not two populations. We split 120 into 36 + 84 on panel 5; we never count 120 + 120 = 240.

ILLUSTRATIVE: three rows of bg_2020_cells

bg_2020	PUMA5CE	CDFP	vap20
G1	P1	D1	60
G1	P1	D2	140
G2	P1	D1	100

G1, G2, P1, D1 and D2 are invented identifiers, not real Census codes.

ACTUAL: reviewed relationship and identifier-query results

bg_2020	candidate bg_cvap	land_m2	ID returned?
090093431011	091403431011	3,606,413	Yes
090093431011	091703431011	160	No

The query uses Census TIGERweb’s “Census Block Groups” geographic layer, dated 1 January 2024. It returns identifiers and area attributes, not CVAP counts.

ILLUSTRATIVE: temporary joined rows, before contributions

bg_cvap	CDFP	vap20	repeated cvap_est
G1	D1	60	120
G1	D2	140	120
G2	D1	100	80

The exported contribution table stores the allocated amounts instead of repeated source totals.

Outside Connecticut, the block-group identifiers match directly, apart from the 39 missing New York CVAP records.

Some Connecticut water links describe possible locations rather than an exact partition. Their 2020 population is zero.

Check matches and missing values before allocating

Apply the first matching condition. These are conditions on a block group, not successive transformations of every row.

1 A CVAP group has no matching geography

The CVAP table contains a block-group ID, but our reconciled block table has no matching blocks or PUMA–district portions for it. The run ends with an error for review: it cannot locate this population. No such unresolved case remained in the validated run.

→

ILLUSTRATIVE: a failed and a successful geographic match

bg_cvap	cvap_est	intersections	result
G_missing	120	NA	Error
G_matched	120	2	Continue

2 The geography exists, but CVAP is missing

Keep missing CVAP as NA, never zero. The 39 reviewed New York omissions match the Census suppression list. Their two PUMAs lie entirely in NY-02, so geographic containment can still give a district probability of one.

If the missing-ID set differs from those 39 reviewed IDs, the run ends with an error for investigation.

→

ILLUSTRATIVE IDs: allowed versus unreviewed missing CVAP

bg_cvap	cvap_est	reviewed omission?	contribution
NY_example	NA	Yes	NA
New_example	NA	No	Run errors

3 The published CVAP estimate is zero

All mapped portions receive a contribution of zero because there is no estimated CVAP to distribute. The source estimate is not changed. This rule covered 1,798 block groups. A published estimate of zero is still an estimate, not proof that no citizen adult lives there.

→

ILLUSTRATIVE: one zero-estimate group spanning two districts

bg_cvap	CDFP	cvap_est	cvap_contribution
G_zero	D1	0	0
G_zero	D2	0	0

What “stop” means in practice

The code raises an error and does not reach the final crosswalk save for that run. Review the named audit file, resolve the mismatch or document a new decision, then rerun. Files saved by an earlier successful run may still exist.

Turn each usable block-group total into contributions

A contribution is an estimated number of citizen adults attributed to a PUMA–district portion; it is not yet a probability.

4 All mapped blocks have district code ZZ

There is no assigned district to receive the estimate. The Connecticut water group has one published CVAP estimate of 5, although it is linked to two old block groups and two PUMAs. Keep the published 5 in the audit and record 5 as excluded once; both portion contributions are zero.

ACTUAL: the reviewed Connecticut water group

bg_cvap	PUMA5CE	CDFP	cvap_contribution
091309901000	20500	ZZ	0
091309901000	20401	ZZ	0

ZZ means “no district assigned”. It is a location code, not a population value.

5 The group lies in one assigned portion

The entire estimate goes to that PUMA and district; no population split is needed. This applied to 231,665 block groups.

Connecticut group 091902114003 has CVAP 5. Both original blocks are in PUMA 20904 and CT-05, so all 5 contribute there despite the zero 2020 adult count.

First row ACTUAL; G2 and G3 are ILLUSTRATIVE

bg_cvap	PUMA5CE	CDFP	cvap_est	contribution
091902114003	20904	05	5	5
G2	P1	D1	80	80
G3	P1	D2	100	100

6 Split the estimate using adult shares

If the group spans portions, has a positive 2020 adult total, and no adults in ZZ portions, distribute its CVAP estimate in proportion to those adult counts.

G1 has 200 adults in 2020: 60 in D1 and 140 in D2. Its CVAP estimate is 120. We attribute 120 × 60/200 = 36 to D1 and 120 × 140/200 = 84 to D2. The pieces add back to 120. This rule covered 6,273 groups.

ILLUSTRATIVE: one group divided across two district portions

bg_cvap	CDFP	vap20	adult share	contribution
G1	D1	60	60 / 200	36
G1	D2	140	140 / 200	84
Total	—	200	1	120

The approximation in rule 6

Within a block group, we assume citizen adults follow the same spatial proportions as all adults counted in 2020.

We sum block adult counts into portions first. No observed block-level CVAP counts are available or created.

The reviewed zero overrides: same arithmetic, different evidence

For five groups, geography alone and 2020 adult counts both leave the split across districts unknown.

7 Five reviewed groups cannot be split

These groups span several assigned districts but have zero 2020 population and adults. Rule 6 cannot split their positive CVAP estimates: its denominator is zero.

Our agreed decision sets only these five allocation inputs to zero, reducing the amount allocated by 24. The published estimates stay unchanged. This is a project assumption, not a Census correction or proof of true zero population.

ACTUAL: the complete list of approved overrides

bg_cvap	cvap_est	vap20	allocation input
060379800031	5	0	0
120219900000	4	0	0
120719800001	5	0	0
120869806001	5	0	0
120999900000	5	0	0

All three zero-input routes already converge

The code already uses the same calculation in all three cases: contribution = 0.

Rule 3 starts with a published zero. Rule 7 adjusts a positive estimate. Rule 4 excludes a positive estimate because no district is assigned. The separate labels preserve why the input is zero; the arithmetic is identical.

ACTUAL totals by reason; Input = Published − Adjustment − Excluded

Reason	Published	Adjustment	Excluded	Input
Published zero	0	0	0	0
Five overrides	24	24	0	0
CT water exclusion	5	0	5	0

Display headings abbreviate cvap_est, cvap_structural_zero_adjustment, cvap_geographic_exclusion and cvap_for_allocation.

How the rule works at a zero denominator

The sequence is: allocate directly when geography fixes the destination; otherwise use adult shares if possible; otherwise review. Setting all zero-adult cases to zero CVAP would also remove Connecticut’s retained land estimate of 5.
New unresolved cases end the run. The five overrides apply only if their checked data still match.

Decision summary: geography determines what can be done

Positive CVAP; 2020 adults = 0	Decision
One assigned portion	Allocate all directly
Several assigned portions	Review; five approved zeros
Only unassigned portions	Exclude; retain source

A previously unseen exception ends the run. The allocation decision record identifies all five approved overrides.

Known published CVAP: 240,785,129 = 240,785,100 contributed + 24 adjustment + 5 exclusion. NY unknowns are omitted.

Add the contributions, then calculate district probabilities

The final result has one row per state / PUMA / assigned district, with a probability between zero and one.

Add citizen-adult contributions within a PUMA

For each district, add contributions from all block groups in the PUMA. In our example, D1 receives 36 + 80 = 116 and D2 receives 84 + 100 = 184. The PUMA total is 300. If any contribution is unknown, its complete total stays unknown.

ILLUSTRATIVE: totals from G1, G2 and G3

PUMA5CE	CDFP	district_cvap	puma_cvap
P1	D1	116	300
P1	D2	184	300

FIRST: is the whole PUMA in one district?

Check every original block, including zero-population blocks. If all have the same assigned district and none has code ZZ, give that district probability 1. This handles 1,110 PUMAs, including the two NY PUMAs with suppressed CVAP. Their unknown population totals remain unknown.

ACTUAL: New York containment results

STATEFP	PUMA5CE	CDFP	puma_cvap	allocation_weight
36	03310	02	NA	1
36	03313	02	NA	1

OTHERWISE: divide by a complete PUMA total

Require no missing contributions and a finite, positive PUMA total. Divide each district total by that PUMA total. This handles 1,352 PUMAs. If the required total is missing or nonpositive, the run errors for review.
Here: P(D1 | P1) = 116/300; P(D2 | P1) = 184/300.

ILLUSTRATIVE: probabilities rounded here only for display

PUMA5CE	CDFP	district_cvap	allocation_weight
P1	D1	116	0.386667
P1	D2	184	0.613333

Save the checked probability table

Each state–PUMA–district key must be unique. Probabilities must lie from 0 to 1 and sum to 1 within each PUMA, allowing a numerical tolerance of 10^{-12}. Keep zero-weight intersections; remove ZZ from the final table. The output is puma_cd_crosswalk.rds.

ACTUAL: two rows of the saved crosswalk

STATEFP	PUMA5CE	CDFP	allocation_weight	weight_method
36	03310	02	1	structural_single_district
36	03313	02	1	structural_single_district

Full table: 2,462 PUMAs; 435 House districts + DC; 4,183 rows, including 109 zero weights.

These are PUMA-level probabilities. Every person in the same PUMA receives the same distribution when it is applied.